

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Mixed

10 questions · ~15 min

02

A right circular cylinder has a height of 8 meters m and a base with a radius of 12 m. What is the volume, in m^3 , of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

The equation $7g + 7b = 840$ represents the number of blue tiles, b , and the number of green tiles, g , an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

ALGEBRA

$$5x + 3y = 38$$

$$x + 3y = 10$$

In the solution (x, y) to the system of equations above, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

ALGEBRA

In the xy -plane, the graph of the linear function f contains the points 0, 3 and 7, 31. Which equation defines f , where $y = fx$?

- A. $fx = 28x + 34$
- B. $fx = 3x + 38$
- C. $fx = 4x + 3$
- D. $fx = 7x + 3$

Q6

ADVANCED MATH

A rectangle has a length of x units and a width of $x - 15$ units. If the rectangle has an area of 76 square units, what is the value of x ?

- A. 4
- B. 19
- C. 23
- D. 76

Q7

ADVANCED MATH

$$\sqrt[3]{x^3y^6}$$

Which of the following expressions is equivalent to the expression above?

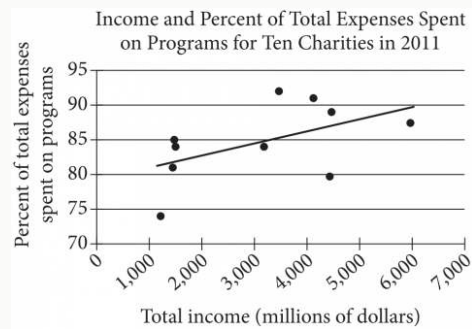
- A. y^2
- B. xy^2
- C. y^3
- D. xy^3

Q8

ADVANCED MATH

A rectangle has a length that is 15 times its width. The function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet



The scatterplot above shows data for ten charities along with the line of best fit. For the charity with the greatest percent of total expenses spent on programs, which of the following is closest to the difference of the actual percent and the percent predicted by the line of best fit?

- A. 10%
- B. 7%
- C. 4%
- D. 1%

From a population of 50,000 people, 1,000 were chosen at random and surveyed about a proposed piece of legislation. Based on the survey, it is estimated that 35% of people in the population support the legislation, with an associated margin of error of 3%. Based on these results, which of the following is a plausible value for the total number of people in the population who support the proposed legislation?

- A. 350
- B. 650
- C. 16,750
- D. 31,750